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IMAGE COMPOSITION USING DEEP LEARNING FOR PRODUCT DESIGN ENGINEERING

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Final Thesis Report

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DEDICATION

Thank you, my wife and my family members, for supporting me in the journey.

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Pramod Vishwanath Pandit

ABSTRACT

According to the survey, most of the data science techniques will get absorbed in all industries in some form by 2025. In the present era of 4th data science evolution, machine learning has started enormously getting used in multiple industries in multiple applications like the energy sector, medical, agriculture, retailing, manufacturing, textile, and cybersecurity. One of the core applications in the field is image processing which is part of deep learning where someone can not only create artificial images but also further refine their clarity and naturality. On another hand, all engineering branches must use 2D images in some forms in their business operations like proposals for newly created concepts, power presentations to clients or business partners, and engineering drawings by their designers not limited to mechanical, production, industrial, civil, and architecture branches but many more. However, there is still a huge gap between the market demand due to emerging technologies and new product development duration. To link artificial intelligence and all traditional engineering fields, deep learning technique enhancement, and its implementations in that field are the keys to filling up this gap and business progression. The product development time depends on the engineer's creative skills, design-development tools and their Product release cycle, and flexibility for the volatility of the market. Here creating artificial product images for new concepts can save predominantly pre-screening or concept selection steps in the Product development cycle. Aligned to that in deep learning presently models used for compositing images are VGG19-based and Tensor flow hub-based magenta, arbitrary-image-stylization-v1-256. Both methods were found promising. However, the magenta was del found better in terms of image quality as per the result image expectations and that is with lesser time consumed. So, recommending magnet model for such a product designing process.

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LIST OF ABBREVIATIONS

AI.....	Artificial intelligence
ML.....	Machine learning
DL.....	Deep learning
GRR.....	Grand review research
GAN.....	Generative adversarial network
L-BFGS.....	Limited-memory Broyden–Fletcher–Goldfarb–Shanno algorithm
CAD.....	Computer-Aided Design
SOPA.....	Slow object placement assessment
FOPA.....	Fast object placement assessment
VGG.....	Visual Geometry Group
CPU.....	Central processing unit
GPU.....	Graphics processing unit

CHAPTER I

INTRODUCTION

1.1 Background

Nowadays, most data science tools like machine learning are getting used in all industries in some form like data prediction or data segmentation. On other hand, to satisfy market demand, companies can reduce their new product development period. Presently the product development lifecycle is understanding customer needs, exploring new ideas, design evaluation, and release in the market. Design evaluation and market release processes are generally well set as per the organization's standards. The major time-consuming is understanding customer needs and more than exploring new ideas. On average designers may take 1 week to 1 month for new concept generation as per the market trends before sharing to leadership for the pre-screening or initial concept selection. It involves majorly Computer-Aided Design (CAD) activities but more than that understanding the current market trends. Both activities can be automated at least for pre-screening by using a deep learning algorithm. After the pre-screening stage, detailed designing can be initiated by designers. Such automation techniques may help organizations to save their initial product design cycle period.

1.2 Problem Statement

Likely said, in any organization of mechanical, civil, and architectural designers, designers have to invest a considerable period of available product designs from the internet or in the competitive market. On average this research may take 1 day to 1 week. After benchmarked product or child part selection, the designer starts a CAD task to design the company's final product assembly with the benchmarked part or sub-assembly. This activity again takes a considerable period from 1 week to 1 month.

So, on average 1 to 1.5 months just go into just concept pre-screening with the company's leadership. Again, the possibility of back and through in concept selection from peers or changes in end-customer demand may stretch the product designing stage up to 2 months. Hence there is an extreme necessity for some automation that drastically reduce the design period to retain customer and hence the business.

An alternate untrusted time-consuming manual process would be cropping required profiles from benchmarked images of required design and simply pasting them on the target image of final assembly in PowerPoint or Photoshop editor. However, it is very subjective for

organizations and its performance depends on the operator's skills and tools quality, and availability.

On another hand, due to upgraded machine learning and deep learning methodologies like artificial image creation, image processing, image segmentation, image blending, and style transfer using selective Generative adversarial neural networks such automation at pre-screening can be expedited at the initial designing stage.

It would not only give consistent good image quality through algorithms but also in lesser time probably in a couple of minutes which directly saves designers' efforts. It would improve organizations' effective resource utilization and productivity.

1.3. Research Questions

Questions on research-

- 1) What % of time designer invest in concept creation?
- 2) How deep learning can help in concept creation?
- 3) What methodology to apply for pre-processing the collected image data?
- 4) Which deep learning algorithm is suitable for image blending and image processing?
- 5) To what extent Computer vision can help in the Product design process?

1.4 Aim and Objectives

The organization named "VIVA RAILINGS" is a modular system manufacturer firm in Texas, United States where the Mechanical Product design team of New Product development works for a project named "DIY" (Do it yourself).

The team aims to design and develop new products or upgrade existing VIVA family products as per current market trends & designs. To understand and collect various designs as inputs for this project the resources engineering folks are using in digital marketing cover competitive designs available in the online market.

Design engineers would like to blend the latest designs with current products to make new VIVA designs.

Aiming the study into consideration the research objectives are:

- To collect the image data of current competitive railings from the internet or competitor's websites.
- To optimize collected image data.
- To implement the deep learning models for collected data.

- To evaluate the performance of the different image blending models used.

1.5 Significance of the Study

This study will help to understand and check the possibility of artificial images created by deep learning algorithms will it help any organization in designing any product and to what extent it can help?

If this method gets proven then other industrial Product designers and architects can use such algorithms to speed up their design process to develop their products from artificial but realistic images.

Generally, for Product based companies 30-40% of Product lifecycle time is consumed by the new product development phase which is considerable. Also, any changes in project restarts the development phase which is again the loss of companies of efforts taken, money, and resources engaged. Hence there is a need for a faster and more accurate product design process right at the first time.

1.6 Scope of the Study

Style and new design images would be taken from Amazon digital marketing websites or VIVA railings competitor's websites. And benchmarked product images would be taken from VIVA railing's current products database.

Considering the time constraints and thesis's limited scope to product design, deeper image comparison parameters image contrast level, image light colour index, photorealism scores, and style faithfulness preference is not considered in this study.

CHAPTER II

LITERATURE REVIEW

2.1 Introduction

As aforementioned, product design engineering is one of the key processes of any product-based company. Design engineers must invest predominant time in new concept generation which may take 1 week to a couple of months depending upon the product type. Last 2 – 3 decades after growing information technologies like the internet & data available on it have. So, through the marketing team's research, designers can get some clues the about expected new products shape.

2.2 Recommended system

As a part of the literature study there are two broad-level aspects. The first focus is on the current challenge which designer faces while creative work & need of faster creative work it may be human created or artificially created. Second aspect is what are current AI driven ways to create artificial images which can assist the designer in new product design process which takes care of aesthetic aspects of the product especially.

This thesis emphasis to connect creative alternate way in the new product design domain to AL-ML emerging technologies.

2.2.1 Current challenges in creative work

Generally, designers while creating any new design, are influenced by existing products or there may be some innovative approach that does not exist before. Likely said, by Hong-Sen Yan in his book Creative Design of Mechanical devices, *“Experience has shown that an engineering designer spends more than the time slowed to arrive at the first design concept. If the process of creative design can be better organized, made systematic and even automatic, more innovative designs may be invented (Yan, 1998).”*

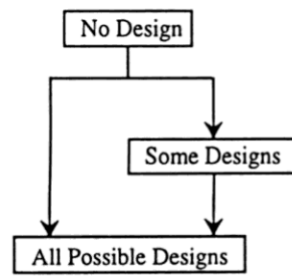


Fig 2.1 Types of creative design of mechanical devices (Yan, 1998)

As shown in the above figure, the new product of all possible designs can be created from existing benchmarked current product with some designs or can be created with some unique innovative idea for which there is no earlier design exists means no design.

“Client and customers usually require improvements rather than novelties (Yan, 1998).”

Hence designers generally select similar designs available while creating new designs.

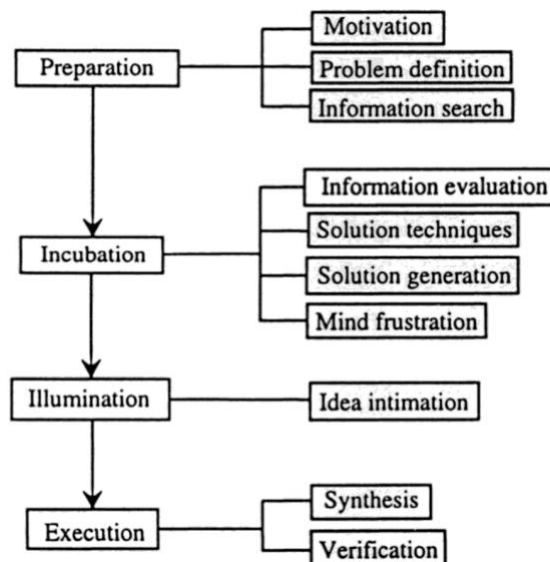


Fig 2.2 Creative Process (Yan, 1998)

The above figure shows various steps involved in any new creating process i.e., Preparation or new creation activity, Incubation or concept development, Illumination or spreading of new creativity or acceptance by people, and execution or implementation in respective fields.

2.2.2 Incubation Phase

“Once the problem has been defined and sufficient information has been obtained, the next stage of the creative process is incubation. This phase is critical to creative solutions. It is usually a long and lonely process, and it is a period for allowing one’s subconscious to work. The incubation phase may include steps of information evaluation, solution techniques, solution generation, and mental frustration (Yan, 1998).”

2.2.3 Current practice to create a new concept

Brainstorming session

“Brainstorming is an individual or group method for generating ideas, increasing creative efficacy, or finding solutions to problems (Sutton R, 1996).” Sometimes it works well. However, it is not the perfect way. *“Experimental research indicates that people in face-to-face brainstorming meetings are less efficient at generating ideas than when working alone. This so-called productivity loss has led many brainstorming researchers to conclude that there is overwhelming evidence for the ineffectiveness of these sessions (Sutton R, 1996).”*

Work in isolation or with team members

Most designers follow it by working in isolation and maintaining non-socialness. Its effectiveness is subjective as it depends on one individual's creative skillset, data gathered and concept designing tools used e.g., 3D modelling software, PowerPoint, or photoshop editor. Working with the team members can also assist however it will eat other team members working hours too.

New concept creation by using a deep learning algorithm

As aforementioned, digital marketing, animation, photography, medicine, and many more fields are exploring their image composition applications using deep learning algorithms.

“Image composition is an important operation to create visual content. Among image composition tasks, image blending aims to seamlessly blend an object from a source image onto a target image with light mask adjustment. A popular approach is Poison image blending, which enforces the gradient domain smoothness in the composite image (Zhang L, 2019).”

2.3 Challenges in the recommender system

To implement AI-ML technologies in product design, need to study what are other researcher did till now in image processing. From all listed supportive references below are some kinds of lite which found close for the railing applications.

Deep image blending (Zhang L, 2019)

In this study, two image processing methods were merged. The first step of Poison's image blending is for blending an object from source image to target image. The limitation of individual Poison's image blending is it will only consider the boundary pixel of target image

and this cannot adapt texture of the target image. The second stage was style transfer from style image on content image using VGG16 CNN and iterations using L-BFGS solver. Results proved that it not only gives better blended boundaries but also consistent texture into the blended regions. Hence it is a strong baseline a state-of-art approach for paintings and real-world images.

Making Images Real Again: A Comprehensive Survey on Deep Image Composition (Cong W, 2022)

In this study, author worked on multiple images composing subtasks like image editing using masks for image segmentation, object placement, image blending, image harmonization and shadow generation. For the image placement method author used various methods like TERSE, PlaceNet, SOPA, and FOPA.

On another side, even if the author succeeds to prove it at a satisfactory level, correlating to the product design process, the basic issue is image pre-processing i.e., cropping the initial unknown image got from digital marketing websites as per the required boundary. A conventional method of such cropping and masking is done on basis of pretrained image database like VGG which have earlier similar category images. If this data is not available like railing subparts and their child parts images then image cropping becomes challenging and again needs to go back to a manual editing process like MS paint or MS PowerPoint to create any content image.

An Overview of Color Transfer and Style Transfer for Images and Videos (S, 2022)

In this study, author focused on selective region's color transfer and style transfer for images using a pre-trained VGG19 model. Here content and style images are very close and similar in pattern hence color and style transfer smoothly. However, this is not always possible as content and style images are different from each other like in railing system images.

Deep Photo Style Transfer (Luan F, 2017)

In this study, the author added features of photorealism regularization and augmented style loss with semantic segmentation in traditional neural style transfer, it is controlled by number the of convolutional layers, the weight control to style loss and weight to configure layer preference, and weight that controls photorealism regularization. Benefit from this model is it succeed to suppress distortion and yields from generated image to make it more photorealistic. However, it's supported by VGG19 pre-trained model with NVIDIA Titan X GPU.

2.4 Discussion

Reviewing the literature, below are some gaps observed from product design timeline perspective.

- There is a need that designer must expedite the creative activities at least the pre-screening or concept phase which is the most time-consuming in any product lifecycle.
- The time has come to discover and take support in the design process using some advanced AL-ML evolved technologies.

Also below are some gaps observed in current image style transfer methodology perspective to correlate with product design process.

- In general, all style transfer methods using some sort of training data in a database as images captured either all over world of living or non-living things like people, trees, artist and many more. But from any organization's perspective such training data of product images may or may not be available. So, it is difficult to effectively use AI ML methods. There are exceptions like medical X-ray images which is again confidential and domain perspective data.

2.5 Conclusion

Research on deep image blending by (Lingzhi Zhang, Tarmily Wen and n, and Jianbo Shi, 2019) (Zhang L, 2019) is considered a reference for this study. And further newly evolved state-of-the-art algorithms for image composition are implemented in this study. There is more advancement in state-of-the-art algorithms for image blending for other applications like medical or digital marketing but less useful in the product design engineering process. So, taking the opportunity to explore how Viva railing organization's products can be designed in form of concept images which will be created artificially by deep learning algorithms.

CHAPTER III

RESEARCH METHODOLOGY

3.1 Introduction

This study follows the methodology with a step-by-step process.

- The First step will be the collection of competitive railings image data. It will include images of sub-assemblies and child parts of railing systems from various resources from e-commerce websites like Amazon or VIVA competitors' products where marketing team can understand market trends.
- The Second step will be pre-processing the collected image data means converting the data into a format relevant to the study.
- The Third step will be building a model using deep learning for image blending between the target image of VIVA railings products and the style image of market trends.
- The Fourth step will be evaluating the deep learning techniques for the performance using appropriate evaluation measures and comparing different image composition methods or models to select the best model considering the scenario and relevant metrics.

Briefly below would be the methodology pipeline.

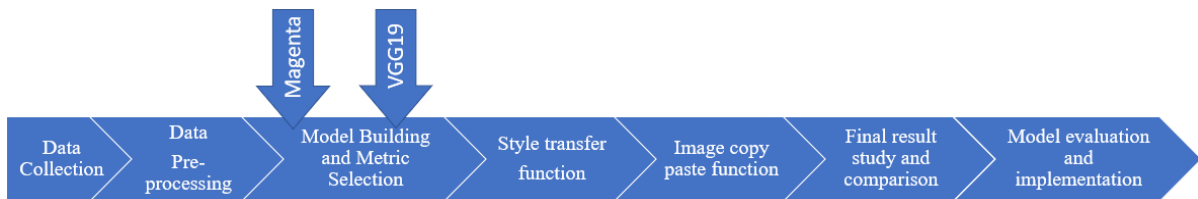


Fig.3.1. Methodology Pipeline

Some specific requirements are as below:

- Since Product data is limited to organization and specific railing products used in the construction and architecture field, probably its training data will not be available.
- Results of generated images will be shared within the organization for voting purposes.

The dataset link is https://github.com/Pramod077/Deep_Learning

The data source size is 13.8MB

3.2 Related work

For image blending the traditional way is the Intelligent Scissors method in Photoshop. However, in the era of AI-ML and deep learning technologies, many researchers have been working on various image composition algorithms or methods or supportive datasets like VGG16, VGG19, Laplacian Pyramid Blending, Graph Cuts in Image Segmentation, Poisson cloning, Gradient domain image editing, Seamless Poisson cloning. Google is also exploring similar algorithms like Magenta.

Considering our project scopes, objectives, and research done on deep image blending by (Lingzhi Zhang, Tarmily Wen, and Jianbo Shi, 2019) (Zhang L, 2019) Poison’s image blending method is more closed and relevant.

3.3 Methodology

3.3.1 Data Collection

Railing product sub-assemblies and child parts' image data will be collected from E-commerce websites.



Fig. 3.3 Data collected from E-commerce websites

As shown in above figure, multiple competitive railing related products available on online marketing resources are searched through internet and their images will be downloaded as per product search.

3.3.2 Data Preparation

Once data is collected it will be manually cropped as per the required shape and orientation to make it ready for use. However, while pre-processing in algorithms, target and style images will be resized as per requirements.



Fig. 3.4 Image pre-processing

As shown in above figure, targeted images may need to edit by using Microsoft paint or Office powerpoint tools.

3.3.3 Model Building and Metric Selection

Once image data is ready, the model building can be initiated. The expected algorithm's basic flow will be as below.

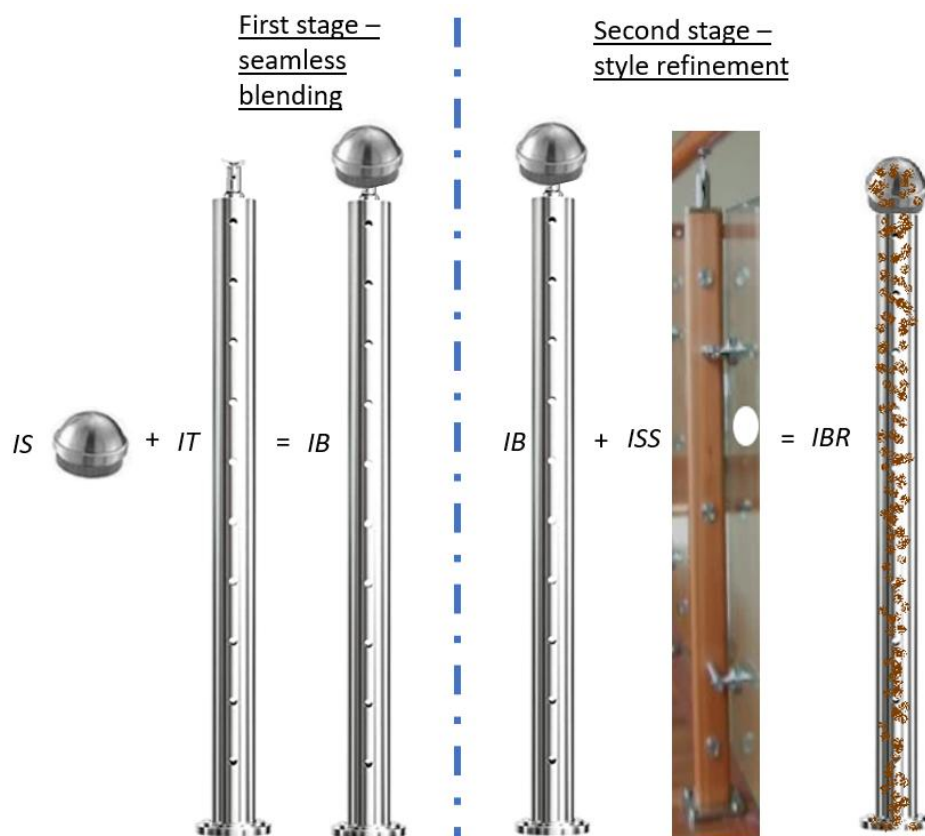


Fig. 3.5 Proposed algorithm process flow

Above figure shows potential process flow of image processing. At first stage simply copy paste of two parts (IS and IT) can be done either using conventional way like MS paint or MS powerpoint or some python libraries like Pillow to create content image (IB) ready for style processing. At second stage content image (IB) and style image (ISS) can be processed to result generated image (IBR) which is the expected end result. However, both first and second stage can be swapped as per requirement and their results whichever gives best.

Basic image categories are as below.

IT = Target image (Here VIVA Railings Products images)

IS = Source image (Here images taken from an E-commerce website and later cropped and edited.)

IB = Blend image (Image processed by Poison's blending algorithm of imaging IT and image IS)

ISS = Source Style image (Here images are taken from either the E-commerce website or VIVA railing products)

IBR = Refined blending image (Further refined IB image by blending it with the target image IT)

Poison's image blending basic algorithm's flow will be as below. (Lingzhi Zhang, Tarmily Wen, Jianbo Shi, 2019). It can be referred to as a VIVA railings product, however, for VIVA Products this algorithm may differ due to product structure and end requirements.

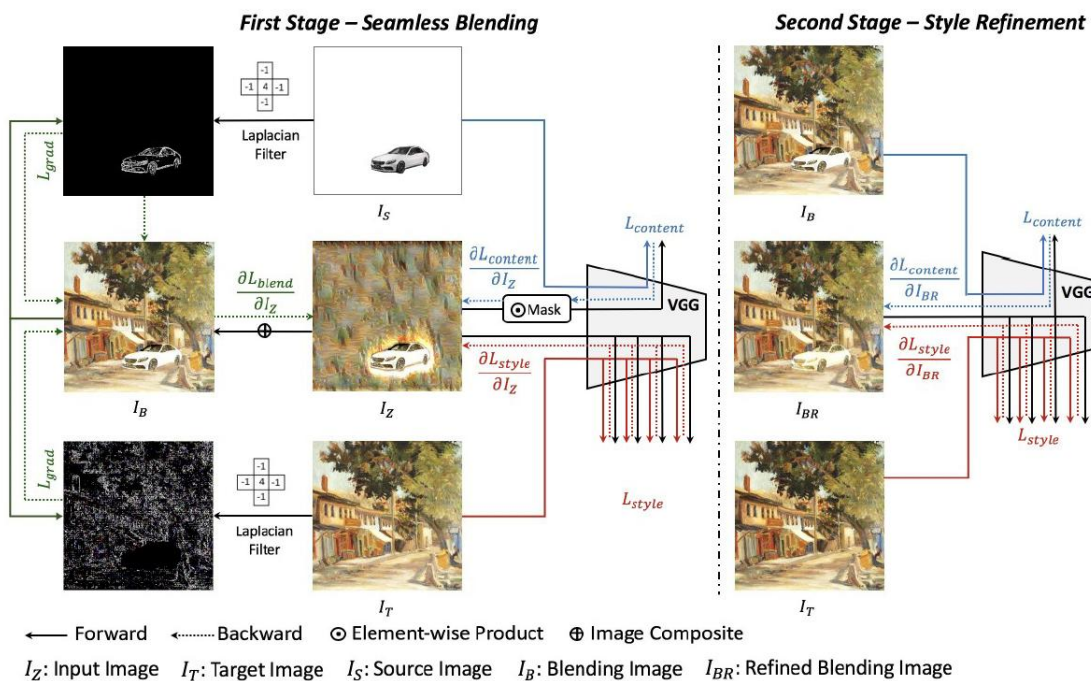


Fig. 3.6 Poison's image blending basic algorithm's flow (Zhang L, 2019)

Algorithm 1 First Stage - Seamless Blending

Input: source image I_S , blending mask M , target image I_T
max iteration T , loss weights $\lambda_{grad}, \lambda_{cont}, \lambda_{style}, \lambda_{hist}, \lambda_{tv}$
Given: a gradient operator ∇ , a pretrained VGG network F
Output: blending image I_B

```

for  $i \in [1:T]$  do
   $I_B = I_Z \odot M + I_T \odot (1 - M)$ 
   $\mathcal{L}_{grad} = \text{GradientLoss}(I_B, I_S, I_T, \nabla)$  by Eq. 6
   $\mathcal{L}_{cont} = \text{ContentLoss}(I_Z, M, I_S, F)$  by Eq. 7
   $\mathcal{L}_{style} = \text{StyleLoss}(I_B, I_T, F)$  by Eq. 8
   $\mathcal{L}_{hist} = \text{HistogramLoss}(I_B, I_T, F)$  by Eq. 11
   $\mathcal{L}_{tv} = \text{TVLoss}(I_B)$  by Eq. 12
   $L_{total} = \lambda_{grad} * \mathcal{L}_{grad} + \lambda_{cont} * \mathcal{L}_{cont} + \lambda_{style} * \mathcal{L}_{style} +$ 
 $\lambda_{hist} * \mathcal{L}_{hist} + \lambda_{tv} * \mathcal{L}_{tv}$ 
   $I_Z \leftarrow \text{L-BFGS\_Solver}(L_{total}, I_Z)$ 
end
 $I_B = I_Z \odot M + I_T \odot (1 - M)$ 

```

Algorithm 2 Second Stage - Style Refinement

Input: blending image I_B , target image I_T
max iteration T , loss weights $\lambda_{cont}, \lambda_{style}, \lambda_{hist}, \lambda_{tv}$
Given: a pretrained VGG network F
Output: refined blending image I_{BR}

```

 $I_{BR} = \text{copy}(I_B)$ 
for  $i \in [1:T]$  do
   $\mathcal{L}_{cont} = \text{ContentLoss}(I_{BR}, I_B, F)$  by Eq. 9
   $\mathcal{L}_{style} = \text{StyleLoss}(I_{BR}, I_T, F)$  by Eq. 10
   $\mathcal{L}_{hist} = \text{HistogramLoss}(I_{BR}, I_T, F)$  by Eq. 11
   $\mathcal{L}_{tv} = \text{TVLoss}(I_{BR})$  by Eq. 12
   $L_{total} = \lambda_{grad} * \mathcal{L}_{grad} + \lambda_{cont} * \mathcal{L}_{cont} + \lambda_{style} * \mathcal{L}_{style} +$ 
 $\lambda_{hist} * \mathcal{L}_{hist} + \lambda_{tv} * \mathcal{L}_{tv}$ 
   $I_{BR} \leftarrow \text{L-BFGS\_Solver}(L_{total}, I_{BR})$ 
end

```

Fig.3.7 Two-stage image blending algorithm process flow (Zhang L, 2019)

As per above Poisson's image blending algorithm, there are two stages. The first stage aims to seamlessly blend the object onto the background, and the second stage aims to further refine the blending region's texture and style. The input to the first stage is a 2D random noise, while the input to the second stage is the final blending image from the first stage. They used the VGG-16 network pretraining on ImageNet to extract features for computing style and content losses. Regarding the gradient domain, they use the Laplacian filter to compute the second-order gradient of images to compute the gradient blending loss. They used VGG layers conv1_2, conv2_2, conv3_3, and conv 4_3 to compute the style loss and conv2_2 to compute content loss. They set the maximum iteration to be 1,000 in both stages, and optimize the loss with an L-BFGS solver. The runtime on a 512 x 512 image takes about 5 minutes on a single NVIDIA GTX1080Ti.

In this thesis case, it needs to exclude i.e. VGG-16 network as there is no need to use the training data of railings images. Since it is a product specific and it's is not available anywhere. Since there just have to create and explore a couple of image blending operations it will exclude the NVIDIA processor usage to make the algorithm simple and able to work with the local computer system without any graphics support like NVIDIA.

About metric selection, in the case of images, the selection of best-blended images is very subjective and varies from person to person. So there are no specific value-driven criteria for model selection.

3.3.4 Model evaluation and implementation

Indeed, any product's aesthetic design using the DL algorithm is an innovative approach. Unfortunately resulting blended images cannot be compared to any existing or new VIVA

RAILINGS products images. Moreover, there is no any ML evaluation parameter for comparing dissimilar blended images. Dissimilar image comparison is very subjective and person-to-person opinions may vary. But here in this case, a human visual check is not an option. And hence Product design team decided to provide a rating as a team to result images generated by different algorithms to compare their results. The rating will be 1 for Good, 2 for Average, and 3 for Bad. Here below railing parts are considered for evaluation.

Case study	Magenta	VGG19	Rating
Post			
Handrail			
Infill			

Table 3.0 Model evaluation matrix

3.4 Tools

Python 3.10.7

Numpy 1.23.3

Pandas 1.5.0

PyTorch 1.12

TensorFlow v2.10.0

Keras 2.10.0

Pillow 9.3.0

CPU Intel Core i5 with Turbo Boosts and above

GPU minimum 8GB in case need to use

4.0 Summary

Deep learning algorithm generally requires large-scale datasets to train the model with hardware requirements with sufficient processing-powered GPUs. However, if the scope of tasks or requirements is not big enough then there can be an easier and cheaper way to exclude nonrequired stuff from algorithms like training data and GPUs.

CHAPTER 4

IMPLEMENTATION

4.1 Introduction

As cited in the literature review there is a need to explore other image-blending algorithms apart from Poison's image-blending method. The reason is to avoid use of costly NVIDIA extra processors. Many products-based companies like VIVA RAILINGS would like to work isolated on data saved on their local systems or local servers. Hence according the requirements limited but state of art ML models shortlisted and approached for further image composition trials.

4.1.1 VGG19

VGG19 is a famous pre-trained CNN model created by ImageNet. It is upgraded from our benchmarked VGG16 model by an increment in number of weight layers from 16 to 19.

4.1.2 Tensorflow Hub Magenta arbitrary image stylization -V1-256

Tensor flow Hub has many pretrained DL models. One of that Magenta which is supported by Google is mostly used for artist applications.

4.2 Dataset

The dataset of product images and all pseudocode are kept on the below path.

https://github.com/Pramod077/Deep_Learning

4.3 Summary

Both VGG19 and Magenta models are implemented on the same datasets. Major outcomes are that Magenta found better in terms of results both for expected blended images and generated image quality. Also, Magenta took lesser time compared to VGG19. Moreover, in one case VGG19 failed to generate an image. So overall it is recommended to use with Magenta model for the VIVA RAILINGS Product design process.

CHAPTER 5

RESULT AND EVALUATION

5.1 Introduction

After implementing both models, the results of railings' the blended image were collected and studied. Likely said earlier, a comparison of results generated by both methods is to be done by ratings like Good = 1, Average = 2, and 3 = Bad.

5.2 Model Output

The table shows the evaluation matrix of the composed railing images for the result evaluation.

Case study (content + style + subpart copy paste by Python Pillow library)	Tensor flow hub-magenta-arbitrary-image-stylization-v1-256 + Copy paste by Python Pillow library	VGG19 + Copy paste by Python Pillow library	Result	Conclusion
Post (Content + Style + Subpart) 	1 = Good 	3 = Bad* 	Magenta is good. VGG19 failed. (*At the 3/100 iteration loss exceed exponentially so the result failed).	Here Magenta method is preferred.
Handrail (Content + Style + Subpart) 	1 = Good 	2 = Average 	Magenta is better than VGG19.	Here Magenta method is preferred.
Infill (Content + Style + Subpart) 	2 = Average 	1 = Good 	VGG19 is better than Magenta.	Here VGG19 method is preferred.

Table 5.0 Model output

As shown above figure, for Post, Handrail and Infill case study Magenta arbitrary image stylization model got better results. In Post case, VGG19 model failed to provide result due increased loss values after iteration 3/100. And hence Magenta got rating (4) compared to VGG19 model (6) where least is better in this case.

5.3 Summary

As per the above result summary, Tensor flow hub-magenta-arbitrary-image-stylization-v1-256 is preferred over the VGG19 method. Also mentioned earlier, after experiments, earlier referred VGG16 based poison's image blending method is excluded in scope due to its

independent way of datasets and training data which does not correlate to VIVA RAILINGS product images.

CHAPTER 6

CONCLUSIONS AND RECOMMENDATIONS

6.1 Introduction

Effective application of state of art DL algorithm by merging with product design engineering is the indeed innovative step. It is inspiring and exciting for many more future opportunities in the Era of data science, AI and ML. This study it is well demonstrated through this thesis. Special chocolate texture creation in handrail design is very impressive as it saved time and with a fresh idea.

6.2 Discussion

In this study, even though targeted components of handrail, infill, and post got good results, some peripheral targeted railing sub-assemblies did not receive the expected results both in magenta and VGG19 method.

- In the magenta method when there are more than two content images, then manual editing is a must. And if it's not done properly, it directly impacts end results. See the image below.



Fig 6.0 Failure in Magenta model

- In the VGG19 method, some style implementations did not get as per expectations. See image below.

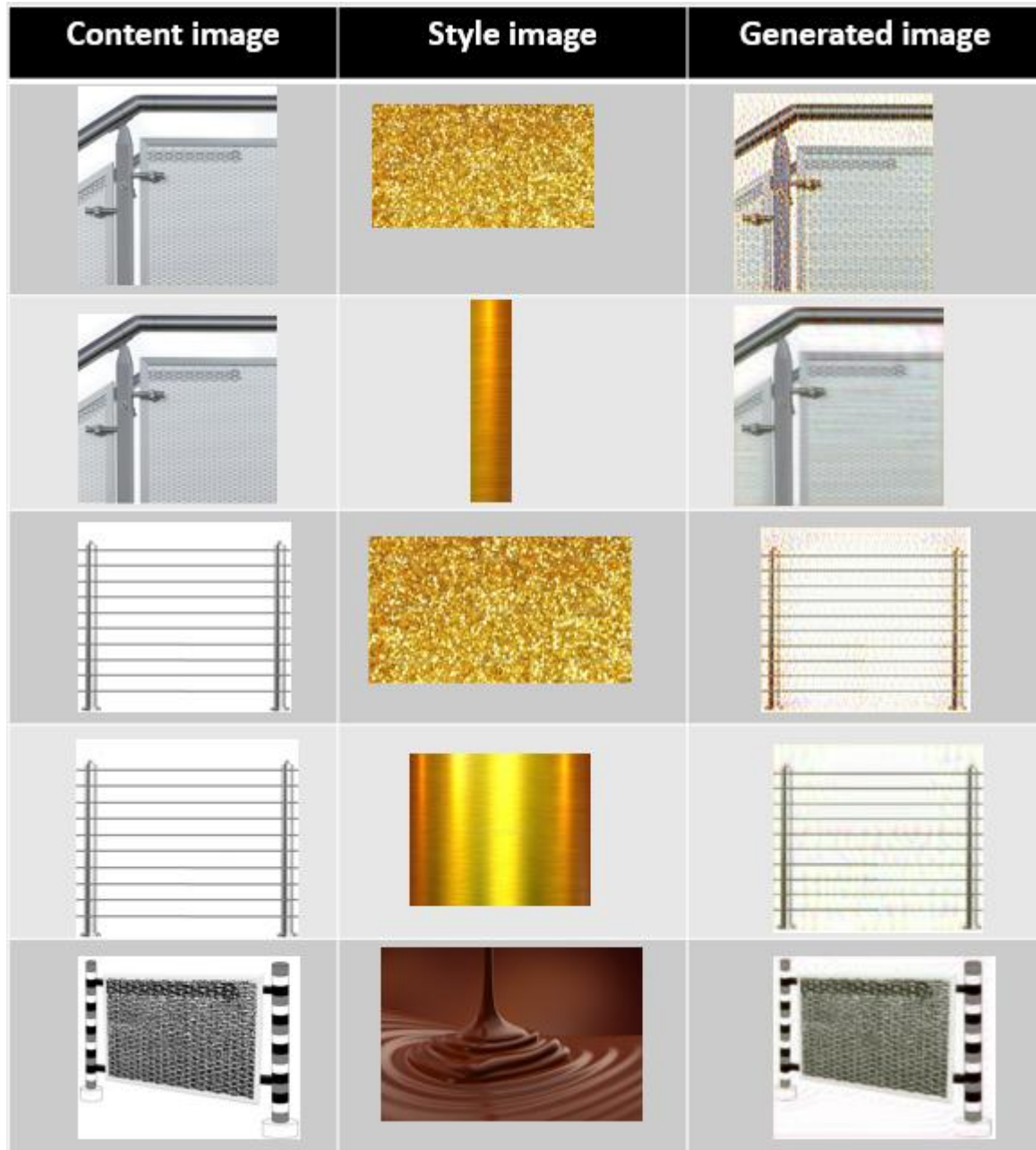


Fig 6.1 Failure in VGG19 model

6.3 Conclusion

There will be always tread off among different image blending methods. Whichever gives the best results can be used as per requirements. There is no perfect image blending method. However, in this thesis case, Tensorflow Hub Magenta arbitrary image stylization -V1-256 is proved better.

6.4 Contribution to the knowledge

1. Image cropping and manual editing can be replaced by other Python Pillow library command however currently only circle & rectangular shape available in it. An exact image cropping algorithm as per product outline is a need.
2. Content weightage and style weightage are driving parameters in the any image blending process. In this specific study in VGG19 below weightage values used which someone case play further until get expected results:

total_variation_weight = 1e-6

style_weight = 1e-6

content_weight = 2.5e-8

6.5 Future recommendations

After this study, it is proved that the DL approach of image blending and image copy-paste can be used in the new product design which would not only save time but also reduce manpower efforts.

Due to lack of time, another image blending method did not include which can be explored further. Also, image cropping as per the product shape algorithm is a need for such projects.

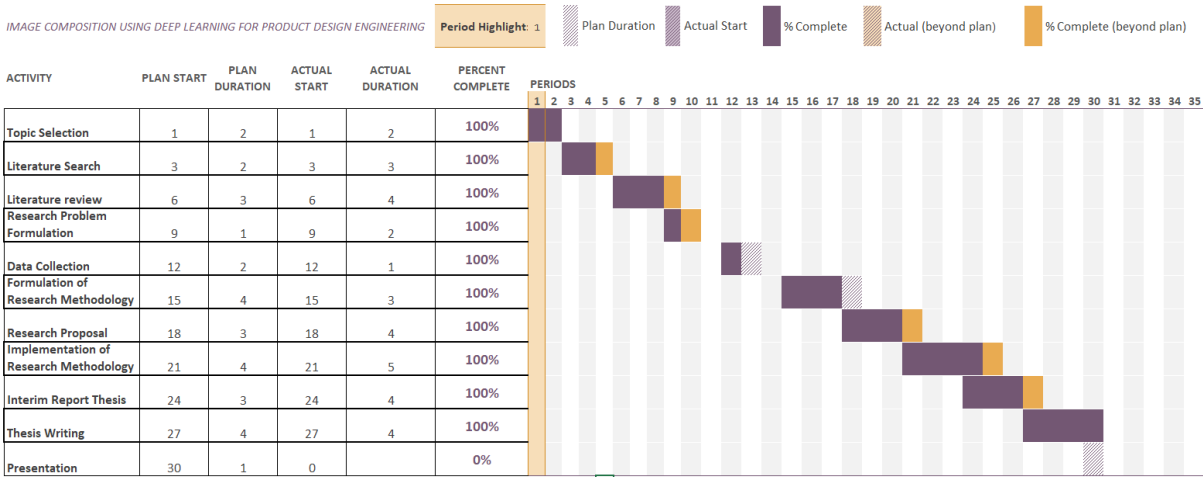
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APPENDIX A: RESEARCH PLAN

Project Planner



APPENDIX B: RESEARCH PROPOSAL

IMAGE COMPOSITION USING DEEP LEARNING FOR PRODUCT DESIGN ENGINEERING

PRAMOD PANDIT

Research Proposal

APRIL 2022

Abstract

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ABSTRACT

As per GRR, the global artificial intelligence market size was valued at USD 93.5 billion in 2021 and is projected to expand at a compound annual growth rate (CAGR) of 38.1% from 2022 to 2030. According to the survey, most of the data science techniques will get absorbed in all industries in some form by 2025. In the present era of 4th data science evolution, machine learning has started enormously getting used in multiple industries in multiple applications like the energy sector, medical, agriculture, retailing, manufacturing, textile, and cybersecurity. One of the core applications in the field is image processing which is part of deep learning where someone can not only create artificial images but also further refine their clarity and naturality. On another hand, all engineering branches must use 2D images in some forms in their business operations like proposals for newly created concepts, power presentations to clients or business partners, and engineering drawings by their designers not limited to mechanical, production, industrial, civil, and architecture branches but many more. However, there is still a huge gap between the market demand due to emerging technologies and new product development duration. To link artificial intelligence and all traditional engineering fields, deep learning technique enhancement, and its implementations in that field are the keys to filling up this gap and business progression. The product development time depends on the engineer's creative skills, design-development tools and their Product release cycle, and flexibility for the volatility of the market. Here creating artificial product images for new concepts can save predominantly pre-screening or concept selection steps in the Product development cycle. Aligned to that in deep learning presently models used for compositing images are VGG19-based and Tensor flow hub-based magenta, arbitrary-image-stylization-v1-256. Both methods were found promising. However, the magenta was del found better in terms of image quality as per the result image expectations and that is with lesser time consumed. So, recommending magnet model for such a product designing process.

Background

Nowadays, most data science tools like machine learning are getting used in all industries in some form like data prediction or data segmentation. On other hand, to satisfy market demand, companies can reduce their new product development period. Presently the product development lifecycle is understanding customer needs, exploring new ideas, design evaluation, and release in the market. Design evaluation and market release processes are generally well set as per the organization's standards. The major time-consuming is understanding customer needs and more than exploring new ideas. On average designers may take 1 week to 1 month for new concept generation as per the market trends before sharing to leadership for the pre-screening or initial concept selection. It involves majorly Computer-Aided Design (CAD) activities but more than that understanding the current market trends. Both activities can be automated at least for pre-screening by using a deep learning algorithm. After the pre-screening stage, detailed designing can be initiated by designers. Such automation techniques may help organizations to save their initial product design cycle period.

Problem Statement

Likely said, in any organization of mechanical, civil, and architectural designers, designers have to invest a considerable period of available product designs from the internet or in the competitive market. On average this research may take 1 day to 1 week. After benchmarked product or child part selection, the designer starts a CAD task to design the company's final product assembly with the benchmarked part or sub-assembly. This activity again takes a considerable period from 1 week to 1 month.

So, on average 1 to 1.5 months just go into just concept pre-screening with the company's leadership. Again, the possibility of back and through in concept selection from peers or changes in end-customer demand may stretch the product designing stage up to 2 months. Hence there is an extreme necessity for some automation that drastically reduce the design period to retain customer and hence the business.

An alternate untrusted time-consuming manual process would be cropping required profiles from benchmarked images of required design and simply pasting them on the target image of final assembly in PowerPoint or Photoshop editor. However, it is very subjective for organizations and its performance depends on the operator's skills and tools quality, and availability.

On another hand, due to upgraded machine learning and deep learning methodologies like artificial image creation, image processing, image segmentation, image blending, and style

transfer using selective Generative adversarial neural networks such automation at pre-screening can be expedited at the initial designing stage.

It would not only give consistent good image quality through algorithms but also in lesser time probably in a couple of minutes which directly saves designers' efforts. It would improve organizations' effective resource utilization and productivity.

Related Research

Introduction

As aforementioned, product design engineering is one of the key processes of any product-based company. Design engineers must invest predominant time in new concept generation which may take 1 week to a couple of months depending upon the product type. Last 2 – 3 decades after growing information technologies like the internet & data available on it have. So, through the marketing team's research, designers can get some clues the about expected new products shape.

Recommended system

As a part of the literature study there are two broad-level aspects. The first focus is on the current challenge which designer faces while creative work & need of faster creative work it may be human created or artificially created. Second aspect is what are current AI driven ways to create artificial images which can assist the designer in new product design process which takes care of aesthetic aspects of the product especially.

This thesis emphasis to connect creative alternate way in the new product design domain to AL-ML emerging technologies.

Current challenges in creative work

Generally, designers while creating any new design, are influenced by existing products or there may be some innovative approach that does not exist before. Likely said, by Hong-Sen Yan in his book Creative Design of Mechanical devices, *"Experience has shown that an engineering designer spends more than the time slowed to arrive at the first design concept. If the process of creative design can be better organized, made systematic and even automatic, more innovative designs may be invented (Yan, 1998)."*

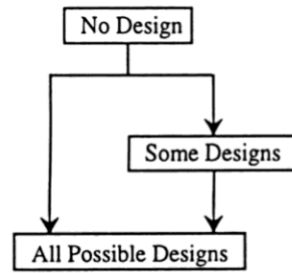


Fig 2.1 Types of creative design of mechanical devices (Yan, 1998)

As shown in the above figure, the new product of all possible designs can be created from existing benchmarked current product with some designs or can be created with some unique innovative idea for which there is no earlier design exists means no design.

“Client and customers usually require improvements rather than novelties (Yan, 1998).”

Hence designers generally select similar designs available while creating new designs.

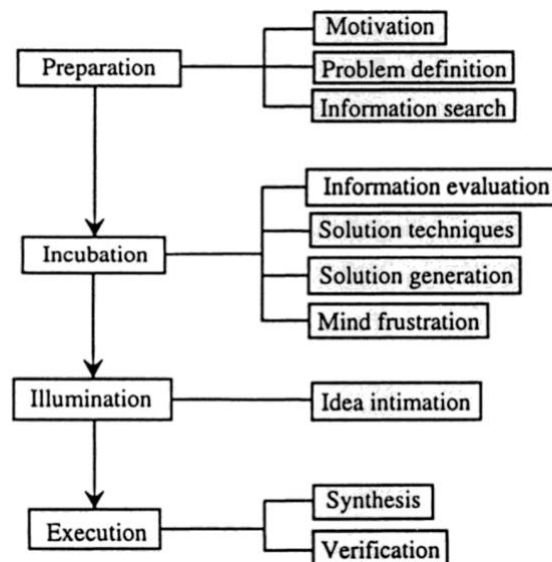


Fig 2.2 Creative Process (Yan, 1998)

The above figure shows various steps involved in any new creating process i.e., Preparation or new creation activity, Incubation or concept development, Illumination or spreading of new creativity or acceptance by people, and execution or implementation in respective fields.

Incubation Phase

“Once the problem has been defined and sufficient information has been obtained, the next stage of the creative process is incubation. This phase is critical to creative solutions. It is usually a long and lonely process, and it is a period for allowing one’s subconscious to work. The incubation phase may include steps of information evaluation, solution techniques, solution generation, and mental frustration (Yan, 1998).”

Current practice to create a new concept

Brainstorming session

“Brainstorming is an individual or group method for generating ideas, increasing creative efficacy, or finding solutions to problems (Sutton R, 1996).” Sometimes it works well. However, it is not the perfect way. *“Experimental research indicates that people in face-to-face brainstorming meetings are less efficient at generating ideas than when working alone. This so-called productivity loss has led many brainstorming researchers to conclude that there is overwhelming evidence for the ineffectiveness of these sessions (Sutton R, 1996).”*

Work in isolation or with team members

Most designers follow it by working in isolation and maintaining non-socialness. Its effectiveness is subjective as it depends on one individual's creative skillset, data gathered and concept designing tools used e.g., 3D modelling software, PowerPoint, or photoshop editor. Working with the team members can also assist however it will eat other team members working hours too.

New concept creation by using a deep learning algorithm

As aforementioned, digital marketing, animation, photography, medicine, and many more fields are exploring their image composition applications using deep learning algorithms.

“Image composition is an important operation to create visual content. Among image composition tasks, image blending aims to seamlessly blend an object from a source image onto a target image with light mask adjustment. A popular approach is Poison image blending, which enforces the gradient domain smoothness in the composite image (Zhang L, 2019).”

Challenges in the recommender system

To implement AI-ML technologies in product design, need to study what are other researcher did till now in image processing. From all listed supportive references below are some kinds of lite which found close for the railing applications.

Deep image blending (Zhang L, 2019)

In this study, two image processing methods were merged. The first step of Poison's image blending is for blending an object from source image to target image. The limitation of individual Poison's image blending is it will only consider the boundary pixel of target image

and this cannot adapt texture of the target image. The second stage was style transfer from style image on content image using VGG16 CNN and iterations using L-BFGS solver. Results proved that it not only gives better blended boundaries but also consistent texture into the blended regions. Hence it is a strong baseline a state-of-art approach for paintings and real-world images.

Making Images Real Again: A Comprehensive Survey on Deep Image Composition (Cong W, 2022)

In this study, author worked on multiple images composing subtasks like image editing using masks for image segmentation, object placement, image blending, image harmonization and shadow generation. For the image placement method author used various methods like TERSE, PlaceNet, SOPA, and FOPA.

On another side, even if the author succeeds to prove it at a satisfactory level, correlating to the product design process, the basic issue is image pre-processing i.e., cropping the initial unknown image got from digital marketing websites as per the required boundary. A conventional method of such cropping and masking is done on basis of pretrained image database like VGG which have earlier similar category images. If this data is not available like railing subparts and their child parts images then image cropping becomes challenging and again needs to go back to a manual editing process like MS paint or MS PowerPoint to create any content image.

An Overview of Color Transfer and Style Transfer for Images and Videos (S, 2022)

In this study, author focused on selective region's color transfer and style transfer for images using a pre-trained VGG19 model. Here content and style images are very close and similar in pattern hence color and style transfer smoothly. However, this is not always possible as content and style images are different from each other like in railing system images.

Deep Photo Style Transfer (Luan F, 2017)

In this study, the author added features of photorealism regularization and augmented style loss with semantic segmentation in traditional neural style transfer, it is controlled by number the of convolutional layers, the weight control to style loss and weight to configure layer preference, and weight that controls photorealism regularization. Benefit from this model is it succeed to suppress distortion and yields from generated image to make it more photorealistic. However, it's supported by VGG19 pre-trained model with NVIDIA Titan X GPU.

Discussion

Reviewing the literature, below are some gaps observed from product design timeline perspective.

- There is a need that designer must expedite the creative activities at least the pre-screening or concept phase which is the most time-consuming in any product lifecycle.
- The time has come to discover and take support in the design process using some advanced AL-ML evolved technologies.

Also below are some gaps observed in current image style transfer methodology perspective to correlate with product design process.

- In general, all style transfer methods using some sort of training data in a database as images captured either all over world of living or non-living things like people, trees, artist and many more. But from any organization's perspective such training data of product images may or may not be available. So, it is difficult to effectively use AI ML methods. There are exceptions like medical X-ray images which is again confidential and domain perspective data.

Conclusion

Research on deep image blending by (Lingzhi Zhang, Tarmily Wen and n, and Jianbo Shi, 2019) (Zhang L, 2019) is considered a reference for this study. And further newly evolved state-of-the-art algorithms for image composition are implemented in this study. There is more advancement in state-of-the-art algorithms for image blending for other applications like medical or digital marketing but less useful in the product design engineering process. So, taking the opportunity to explore how Viva railing organization's products can be designed in form of concept images which will be created artificially by deep learning algorithms.

Research Questions

Questions on research-

- 1) What % of time designer invest in concept creation?
- 2) How deep learning can help in concept creation?
- 3) What methodology to apply for pre-processing the collected image data?
- 4) Which deep learning algorithm is suitable for image blending and image processing?
- 5) Which metrics are to be considered for finding the quality of a processed image?

1.7 Aim and Objectives

The organization named “VIVA RAILINGS” is a modular system manufacturer firm in Texas, United States where the Mechanical Product design team of New Product development works for a project named “DIY” (Do it yourself).

The team aims to design and develop new products or upgrade existing VIVA family products as per current market trends & designs. To understand and collect various designs as inputs for this project the resources engineering folks are using in digital marketing cover competitive designs available in the online market.

Design engineers would like to blend the latest designs with current products to make new VIVA designs.

Aiming the study into consideration the research objectives are:

- To collect the image data of current competitive railings from the internet or competitor’s websites.
- To optimize collected image data.
- To implement the deep learning models for collected data.
- To evaluate the performance of the different image blending models used.

Significance of the Study

This study will help to understand and check the possibility of artificial images created by deep learning algorithms will it help any organization in designing any product and to what extent it can help?

If this method gets proven then other industrial Product designers and architects can use such algorithms to speed up their design process to develop their products from artificial but realistic images.

Generally, for Product based companies 30-40% of Product lifecycle time is consumed by the new product development phase which is considerable. Also, any changes in project restarts the development phase which is again the loss of companies of efforts taken, money, and resources engaged. Hence there is a need for a faster and more accurate product design process right at the first time.

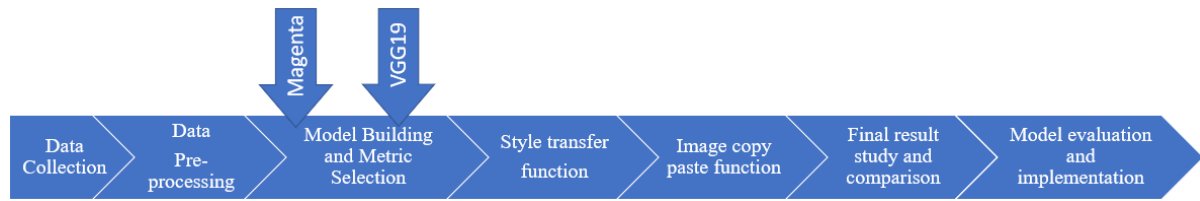
Scope of the Study

Style and new design images would be taken from Amazon digital marketing websites or VIVA railings competitor's websites. And benchmarked product images would be taken from VIVA railing's current products database.

Considering the time constraints and thesis's limited scope to product design, deeper image comparison parameters image contrast level, image light colour index, photorealism scores, and style faithfulness preference is not considered in this study.

Research Methodology Pipeline

Briefly below would be the methodology pipeline.



Data Collection

Railing product sub-assemblies and child parts' images data will be collected from E-commerce websites.



Fig. Data collected from E-commerce websites

Data Preparation

Once data is collected it will be manually cropped as per the required shape and orientation to make it ready for use. However, while pre-processing in algorithms, target and style images will be resized as per requirements.

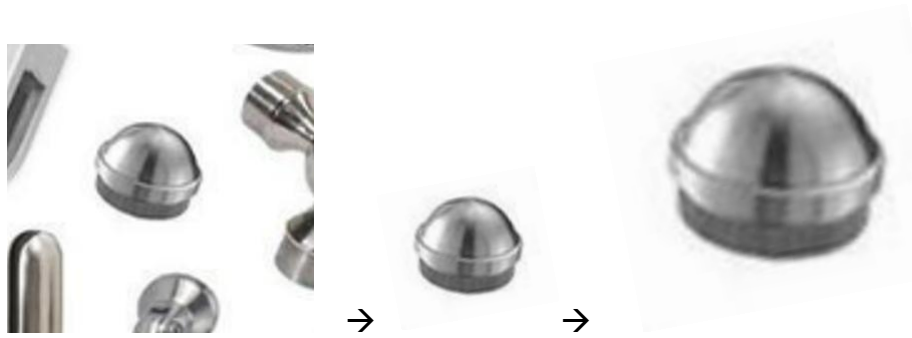


Fig. Image pre-processing

3.3.3 Model Building and Metric Selection

Once image data is ready, the model building can be initiated. The expected algorithm's basic flow will be as below.

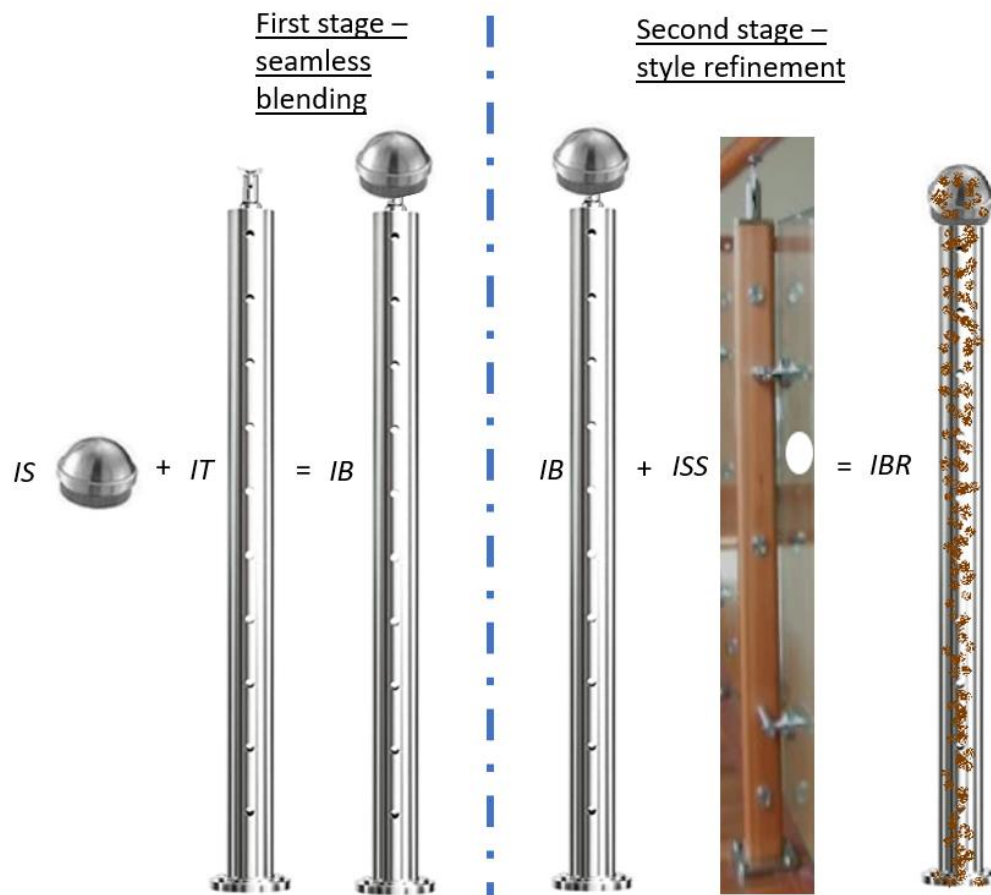


Fig. Proposed algorithm process flow

Basic image categories are as below.

IT = Target image (Here VIVA Railings Products images)

IS = Source image (Here images taken from an E-commerce website and later cropped)

and edited.)

IB = Blend image (Image processed by Poisson's blending algorithm of image I_T and image I_S)

ISS = Source Style image (Here images are taken from either the E-commerce website or VIVA railing products)

IBR = Refined blending image (Further refined IB image by blending it with the target image I_T)

Poisson's image blending basic algorithm's flow will be as below. (Lingzhi Zhang, Tarmily Wen, Jianbo Shi, 2019). It can be referred to as a VIVA railings product, however, for VIVA Products this algorithm may differ due to product structure and end requirements.

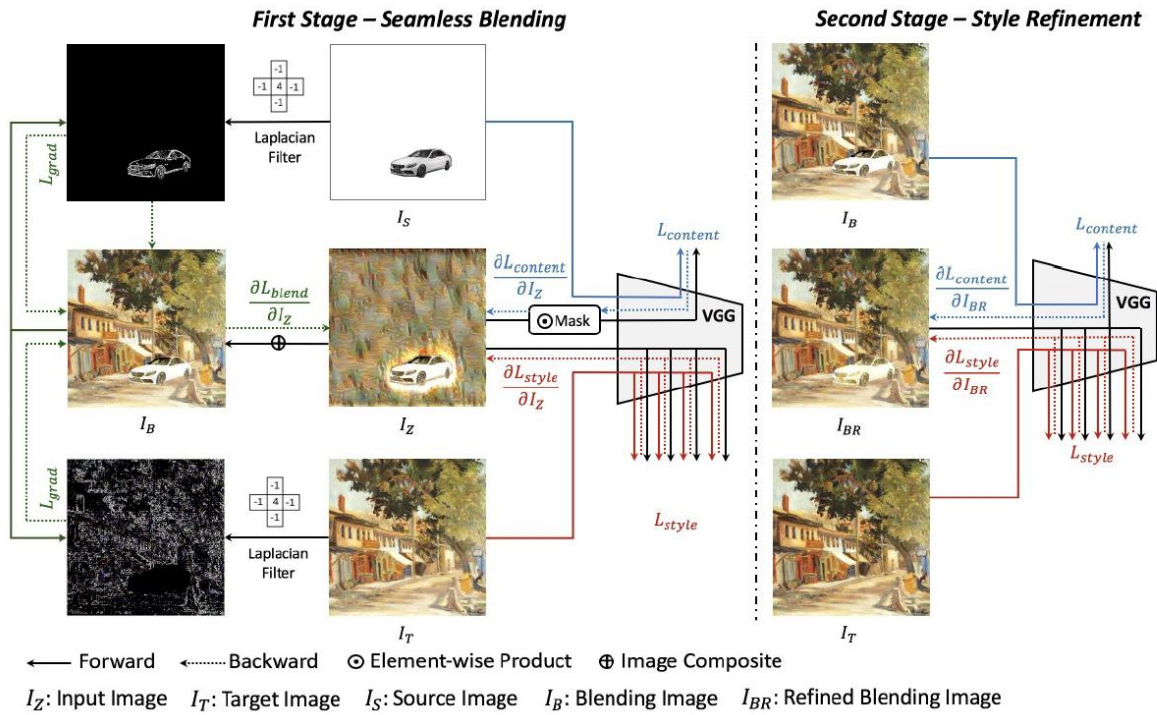


Fig. Poisson's image blending basic algorithm's flow

Algorithm 1 First Stage - Seamless Blending

Input: source image I_S , blending mask M , target image I_T max iteration T , loss weights $\lambda_{grad}, \lambda_{cont}, \lambda_{style}, \lambda_{hist}, \lambda_{tv}$ **Given:** a gradient operator ∇ , a pretrained VGG network F **Output:** blending image I_B

```
for  $i \in [1:T]$  do
   $I_B = I_Z \odot M + I_T \odot (1 - M)$ 
   $\mathcal{L}_{grad} = \text{GradientLoss}(I_B, I_S, I_T, \nabla)$  by Eq. 6
   $\mathcal{L}_{cont} = \text{ContentLoss}(I_Z, M, I_S, F)$  by Eq. 7
   $\mathcal{L}_{style} = \text{StyleLoss}(I_B, I_T, F)$  by Eq. 8
   $\mathcal{L}_{hist} = \text{HistogramLoss}(I_B, I_T, F)$  by Eq. 11
   $\mathcal{L}_{tv} = \text{TVLoss}(I_B)$  by Eq. 12
   $L_{total} = \lambda_{grad} * \mathcal{L}_{grad} + \lambda_{cont} * \mathcal{L}_{cont} + \lambda_{style} * \mathcal{L}_{style} +$ 
     $\lambda_{hist} * \mathcal{L}_{hist} + \lambda_{tv} * \mathcal{L}_{tv}$ 
   $I_Z \leftarrow \text{L-BFGS\_Solver}(L_{total}, I_Z)$ 
end
 $I_B = I_Z \odot M + I_T \odot (1 - M)$ 
```

Algorithm 2 Second Stage - Style Refinement

Input: blending image I_B , target image I_T max iteration T , loss weights $\lambda_{cont}, \lambda_{style}, \lambda_{hist}, \lambda_{tv}$ **Given:** a pretrained VGG network F **Output:** refined blending image I_{BR}

```
 $I_{BR} = \text{copy}(I_B)$ 
for  $i \in [1:T]$  do
   $\mathcal{L}_{cont} = \text{ContentLoss}(I_{BR}, I_B, F)$  by Eq. 9
   $\mathcal{L}_{style} = \text{StyleLoss}(I_{BR}, I_T, F)$  by Eq. 10
   $\mathcal{L}_{hist} = \text{HistogramLoss}(I_{BR}, I_T, F)$  by Eq. 11
   $\mathcal{L}_{tv} = \text{TVLoss}(I_{BR})$  by Eq. 12
   $L_{total} = \lambda_{grad} * \mathcal{L}_{grad} + \lambda_{cont} * \mathcal{L}_{cont} + \lambda_{style} * \mathcal{L}_{style} +$ 
     $\lambda_{hist} * \mathcal{L}_{hist} + \lambda_{tv} * \mathcal{L}_{tv}$ 
   $I_{BR} \leftarrow \text{L-BFGS\_Solver}(L_{total}, I_{BR})$ 
end
```

Fig. Two-stage image blending algorithm process flow

As per above Poisson's image blending algorithm, there are two stages. The first stage aims to seamlessly blend the object onto the background, and the second stage aims to further refine the blending region's texture and style. The input to the first stage is a 2D random noise, while the input to the second stage is the final blending image from the first stage. They used the VGG-16 network pretraining on ImageNet to extract features for computing style and content losses. Regarding the gradient domain, they use the Laplacian filter to compute the second-order gradient of images to compute the gradient blending loss. They used VGG layers conv1_2, conv2_2, conv3_3, and conv 4_3 to compute the style loss and conv2_2 to compute content loss. They set the maximum iteration to be 1,000 in both stages, and optimize the loss with an L-BFGS solver. The runtime on a 512 x 512 image takes about 5 minutes on a single NVIDIA GTX1080Ti.

In our case, some parts have to exclude i.e. VGG-16 network as we won't use the training data of railings images. Since there just have to create and explore a couple of image blending operations we will exclude the NVIDIA processor usage to make the algorithm simple and able to work with the local computer system without any graphics support like NVIDIA.

About metric selection, in the case of images, the selection of best-blended images is very subjective and varies from person to person. So there are no specific value-driven criteria for model selection.

Model evaluation and implementation

Indeed, any product's aesthetic design using the DL algorithm is an innovative approach. Unfortunately resulting blended images cannot be compared to any existing or new VIVA RAILINGS products images. Moreover, there is no any ML evaluation parameter for comparing dissimilar blended images. Dissimilar image comparison is very subjective and person-to-person opinions may vary. But here in this case, a human visual check is not an option. And hence Product design team decided to provide a rating as a team to result images generated by different algorithms to compare their results. The rating will be 1 for Good, 2 for Average, and 3 for Bad. Here below railing parts are considered for evaluation.

Case study	Magenta	VGG19	Rating
Post			
Handrail			
Infill			

Table 3.0 Model evaluation matrix

Tools

Python 3.10.7

Numpy 1.23.3

Pandas 1.5.0

PyTorch 1.12

TensorFlow v2.10.0

Keras 2.10. 0

Pillow 9.3.0

CPU Intel Core i5 with Turbo Boosts and above

GPU minimum 8GB in case need to use

Summary

Deep learning algorithm generally requires large-scale datasets to train the model with hardware requirements with sufficient processing-powered GPUs. However, if the scope of tasks or requirements is not big enough then there can be an easier and cheaper way to exclude nonrequired stuff from algorithms like training data and GPUs.

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APPENDIX A: RESEARCH PLAN

Project Planner

